WEEK 9 · LECTURE L2 · UNIT III

Fairness, Transparency and Bias

How bias enters a model, why fairness definitions conflict, what explanation can and cannot do, and what the DPDP Act does and does not do.

R K Singh · Senior Professor, Department of Commerce · Delhi School of Economics, University of Delhi

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Explain how bias enters a model (Barocas and Selbst; historical, representation, measurement and deployment bias) and how feedback loops feed it.
2. Distinguish demographic parity, equalized odds and calibration, compute them, and explain why they conflict.
3. Distinguish interpretability from explainability and apply Rudin’s argument.
4. State what the DPDP Act 2023 does and does not do for automated HR decisions.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 3 and CLO 5.

COMMIT BEFORE THEORY

Commit to an answer first

A promotion model never uses gender as an input. Can it still disadvantage women?

A

No: without gender as an input it cannot discriminate

B

Yes: other inputs, such as career breaks or part time history, can act as proxies for gender

C

Only if the programmers intended it

D

Only if women are a minority

EXAM LENS

The answer is B.

BAROCAS AND SELBST (2016)

How bias enters a model

Route	How it works	HR example
Defining the target	What “a good employee” means can build in disadvantage	Tenure as the target favors those with fewer career breaks
Labeling examples	Past human judgments used as labels carry past bias	Past hires that were mostly male
Collecting data	Some groups under or poorly represented	Few applications from smaller towns
Selecting features	Features capture group differences rather than merit	Unbroken service years
Proxies	Neutral looking inputs stand in for protected attributes	Career gaps for gender; pin code for caste or class
Masking	Bias introduced deliberately and hidden behind these routes	A chosen proxy disguised as merit

EXAM LENS

Most discrimination by data mining is unintentional, which makes it hard to see and to challenge. The teaching note adds measurement bias (biased ratings as the target) and deployment bias (a model used beyond its purpose).

FEEDBACK LOOPS AND INTERSECTIONS

Bias that feeds itself, and bias at the intersections

The loop

A lower score brings fewer shifts, orders or projects; less work means fewer chances to perform and be rated; the model reads the thinner record as weaker performance; the next allocation is smaller still.

Audit over time

Audit how scores change over time, not only a snapshot: early disadvantage grows unless something outside the loop intervenes.

Buolamwini and Gebru (2018)

Commercial gender classifiers erred up to 34.7% for darker skinned women against 0.8% for lighter skinned men. A model fair by gender and by age can still be unfair to young women: audit combinations, though small groups make estimates unstable.

THREE FAIRNESS DEFINITIONS

Each captures a real intuition, and they pull apart

Demographic parity

Each group is selected or flagged at the same rate (independence; the four fifths rule is a loose relaxed version).

Equalized odds

False positive and false negative rates are equal across groups (Hardt, Price and Srebro, 2016).

Calibration, or predictive parity

A given score means the same chance of the outcome in every group; flags are equally precise.

A WORKED EXAMPLE

One flight risk model, three verdicts

	Group A (200 staff)	Group B (200 staff)
Leavers (base rate)	40 (20%)	30 (15%)
Flagged	100 (50%)	40 (20%)
Flagged who leave (precision)	25 of 100 = 25%	10 of 40 = 25%
Stayers wrongly flagged (false positive rate)	75 of 160 = 47%	30 of 170 = 18%
Leavers caught (true positive rate)	25 of 40 = 63%	10 of 30 = 33%

EXAM LENS

Demographic parity: not met (50% against 20%). Equalized odds: not met (47% against 18%). Predictive parity: met (25% against 25%). Base rates differ, so no imperfect model can satisfy all three at once.

Illustrative figures (Course framing).

KLEINBERG, MULLAINATHAN AND RAGHAVAN (2017); CHOULDECHOVA (2017)

Why they cannot all hold at once

The result

When base rates differ, a model cannot be calibrated and have equal error rates across groups, unless it is perfect. Improving one fairness measure worsens another.

What it means

A fully fair model on every definition is usually impossible; something must be traded.

What HR must do

Choose the definition openly, justify it to those affected, and record the choice.

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Choosing a fairness definition is a value judgment, not a technical setting. The result is mathematical; which fairness matters is an ethical and legal choice.

MITIGATION

Three places to intervene

Preprocessing

Fix the data before training: reweight underrepresented groups, relabel biased targets, augment data.

In processing

Build fairness constraints into training.

Postprocessing

Adjust outputs after training, for example with group specific thresholds.

EXAM LENS

Ask the causal question too: does a protected attribute affect the outcome through legitimate or illegitimate paths?

INTERPRETABILITY OR EXPLAINABILITY?

And Rudin’s argument

Interpretability

How far a person can understand how the model works: simulatability (can you run it in your head?), decomposability (does each part have a meaning?), algorithmic transparency.

Explainability

Post hoc accounts (such as LIME and SHAP) of why a black box produced an output: approximations, which may not match what the model did.

Rudin (2019)

For high stakes decisions, use models interpretable by design rather than explained black boxes; for structured data such as HR records the accuracy cost is often small or nil. Images and free text may still need complex models.

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A model can be explainable and still wrong: explanation is not correction. Levels of explanation run from a bare score, to feature importance, individual explanation, counterfactual (“what would need to change”) and full transparency; transparency should rise with the stakes.

THE DPDP ACT 2023

What it does and does not do for automated decisions

Does not	Does
Regulate algorithms as such	Require accurate, complete data when data are used for decisions
Give an explicit right to an explanation	Require security safeguards and breach reporting
Let consent block staff profiling (employment is a legitimate use, Section 7(i))	Require erasure once the purpose is served, and a grievance route

EXAM LENS

Lawful processing is not the same as fair processing: the Act sets a floor. Audit, human review and a working grievance route are the real safeguards, not consent. Open question: does “employment purposes” cover training models on past staff, or applicants?

APPLY

A promotion model for a bank

Teaching case (Course framing)

A Mumbai private bank builds a model to shortlist officers for promotion. Gender is excluded. Inputs include unbroken years of service, transfers accepted and weekend logins. Women are shortlisted at 60% of the rate for men. The vendor says the model is “fair by design” because it never sees gender.

Work it through

1. Which inputs could act as proxies for gender?
2. Which fairness definition is the vendor implicitly using?
3. Which definition would you choose, and why?
4. Would you accept a black box with explanations? Use Rudin.

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
Remove gender and the model is fair	Proxies carry the protected attribute back in
There is one correct fairness measure	Definitions conflict; choosing one is a value judgment
Explaining a black box is as good as an interpretable model	Explanations can be unfaithful; prefer interpretable models in high stakes uses
Lawful under the DPDP Act means fair	The Act is a floor, not a standard of fairness

COMPARE AND EVALUATE

Six lenses on fairness

Lens	Use it to	Watch out for
Barocas and Selbst (2016)	Name the route of bias	Masking is hard to prove
Hardt et al. (2016)	Define equalized odds	Needs observed outcomes
Kleinberg et al.; Chouldechova	Show the tradeoff	Mathematics cannot choose for you
Buolamwini and Gebru (2018)	Audit intersections	Small groups are unstable
Rudin (2019)	Choose interpretable models	Some data types need complex models
DPDP Act 2023	Set the legal floor	No right to explanation

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (1 of 2)

Using Barocas and Selbst, the impossibility results on fairness, and Rudin’s argument on interpretability, evaluate an Indian IT firm’s plan to use an algorithm to recommend promotions. State which fairness definition you would adopt and why. **20 marks**

A strong answer: identifies three routes of bias; explains the three definitions and why they conflict; adopts and justifies one for this context; applies Rudin to the model choice and notes the DPDP floor.

“There is no such thing as a fair algorithm, only a chosen definition of fairness.” Critically discuss using the impossibility results and an HR example. **20 marks**

A strong answer: defines the three criteria; explains the base rate conflict with numbers; argues the choice is normative; links to accountability.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse (2 of 2)

Evaluate the adequacy of the DPDP Act 2023 for governing algorithmic HR decisions. **20 marks**

A strong answer: states what the Act does and does not do; compares consent based and explanation based approaches; proposes organizational safeguards.

Distinguish explainability from interpretability, with reference to Rudin (2019). **10 marks**

A strong answer: gives precise definitions and Rudin’s argument.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Bias enters through targets, labels, data, features, proxies and loops.**
   Removing the protected attribute is not enough.

2. **Fairness definitions conflict.**
   Choose one openly and defend it.

3. **Prefer interpretable models for high stakes.**
   Explanation is not correction.

4. **The DPDP Act is a floor.**
   Audit and redress, not consent, protect staff.

SET READINGS · FROM THE LESSON PLAN

Read for this week

- Tambe, Cappelli and Yakubovich (2019); Kellogg, Valentine and Christin (2020); Leicht-Deobald et al. (2019).

NOTE

The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Barocas, S., Hardt, M., & Narayanan, A. (2023). *Fairness and machine learning*. MIT Press.
- Barocas, S., & Selbst, A. D. (2016). Big data's disparate impact. *California Law Review, 104*(3), 671 to 732.
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- Chouldechova, A. (2017). Fair prediction with disparate impact. *Big Data, 5*(2), 153 to 163. https://doi.org/10.1089/big.2016.0047
- Government of India. (2023). *The Digital Personal Data Protection Act, 2023*.
- Hardt, M., Price, E., & Srebro, N. (2016). Equality of opportunity in supervised learning. *Advances in Neural Information Processing Systems, 29*.
- Kleinberg, J., Mullainathan, S., & Raghavan, M. (2017). Inherent trade offs in the fair determination of risk scores. *Proceedings of the 8th Innovations in Theoretical Computer Science Conference*.
- Rudin, C. (2019). Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence, 1*(5), 206 to 215. https://doi.org/10.1038/s42256-019-0048-x